Lab 10

Migrating API Products using the API Controller

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs and paths in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal, Admin Portal, Management Console, token endpoint).
- **`<GW_HTTPS_PORT>` / `<GW_HTTP_PORT>`** – API Gateway ports: 8243 (HTTPS) and 8280 (HTTP) plus any port offset.
- **`<PROD_HTTPS_PORT>`** – HTTPS port of the second (production) API Manager: 9445 with port offset 2.
- **`<APIM_HOME>`** – the folder where API Manager is extracted.
- **`<APICTL_DIR>`** – the folder that contains the apictl program after you extract it, for example ~/apictl (Windows: C:\apictl).

The servers use a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue. The -k option in the apictl commands skips the certificate check for the same reason.

The commands use `apictl` and assume its folder is on your PATH (see the Creating an API Project lab). If it is not, run the commands from inside the apictl folder as `./apictl` (Windows: `apictl.exe`).

Training Objective

Learn how to use the API Controller to move an API Product, together with the APIs it uses, between two API Manager environments.

Business Scenario

PizzaShack maintains its API Products in multiple environments. The API Products created in the development environment need to be moved to the production environment smoothly.

High-Level Steps

- Prerequisites: a second (production) API Manager
- Set up the API Controller environments
- Export the CustomerLeasing API Product as a zip file
- Import the API Product into production

Prerequisites

You need two API Manager 4.7.0 instances: the one used in the previous labs as the dev environment (no port offset), and a second one as the production environment with port offset 2. To set up the second one:

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1. Extract a second copy of the API Manager 4.7.0 zip into a different folder.

2. In its `repository/conf/deployment.toml`, add this line under the `[server]` section:

```
offset = 2
```

3. Start it from its `bin` folder. Its portals are on port 9445 and its gateway on 8245 (HTTPS). Running two instances needs about 4 GB more free RAM.

   macOS / Linux

```
sh api-manager.sh
```

   Windows

```
api-manager.bat
```

   If a port is already in use on your machine, use another offset and adjust the URLs (HTTPS port = 9443 + offset).

You also need the CustomerLeasing API Product (and its two APIs) created in the dev environment in the Creating an API Product lab. If you completed the Migrating APIs lab, the production instance and both apictl environments already exist.

Part 1 – Set Up the API Controller

4. apictl must be installed and on your PATH. If you have not done this yet, follow **Part 1 – Install and Initialise apictl** in the Creating an API Project lab. Check it with:

```
apictl version
```

5. List the available commands (optional):

```
apictl --help
```

6. Add the dev environment. Skip this if you added it in the Creating an API Project lab: adding an environment that already exists fails.

```
apictl add env dev --apim https://<APIM_HOST>:<HTTPS_PORT>
```

   Example:

```
apictl add env dev --apim https://localhost:9443
```

7. Add the production environment:

```
apictl add env production --apim https://<APIM_HOST>:<PROD_HTTPS_PORT>
```

   Example:

```
apictl add env production --apim https://localhost:9445
```

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8. List the environments and check that dev and production point to the right ports:

```
apictl get envs
```

Part 2 – Export the API Product from Dev

9. Log in to the dev environment (apictl asks for the password):

```
apictl login dev -u admin -k
```

10. List the API Products in the dev environment. Note the exact **NAME**, **VERSION** and **PROVIDER**:

```
apictl get api-products -e dev -k
```

11. Export the API Product as a zip file. `-n` is the product name exactly as listed (`CustomerLeasing`), not its context: a wrong name fails with `API Revision Not Found`. `-r` is the user who created the product (`publisher` in the Creating an API Product lab). You can use `--latest` instead of `--rev`.

    Command format – macOS / Linux

```
apictl export api-product -n <API-Product-name> -v <version> \
  --rev <revision-number> -r <provider> -e dev -k
```

    Command format – Windows

```
apictl export api-product -n <API-Product-name> -v <version> ^
  --rev <revision-number> -r <provider> -e dev -k
```

    Sample command:

```
apictl export api-product -n CustomerLeasing -v 1.0.0 --rev 1 -r publisher -e dev -k
```

    The zip file is saved in the apictl home folder, for example `~/.wso2apictl/exported/api-products/dev/CustomerLeasing_1.0.0.zip` (Windows: `%USERPROFILE%\.wso2apictl\exported\api-products\dev`).

Part 3 – Import the API Product into Production

12. Log in to the production environment:

```
apictl login production -u admin -k
```

13. Import the API Product. `--import-apis` also creates the APIs that the product uses; without it the import fails with a 404 error when those APIs do not exist in production yet.

    Command format – macOS / Linux

```
apictl import api-product -f <path-to-exported-zip> -e production -k \
  --import-apis --rotate-revision
```

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Command format – Windows

```
apictl import api-product -f <path-to-exported-zip> -e production -k ^
  --import-apis --rotate-revision
```

Sample command:

macOS / Linux

```
apictl import api-product -f dev/CustomerLeasing_1.0.0.zip -e production -k \
  --import-apis --rotate-revision
```

Windows

```
apictl import api-product -f dev/CustomerLeasing_1.0.0.zip -e production -k ^
  --import-apis --rotate-revision
```

Expected output: `Successfully imported API Product`. Check the product and its APIs with:

```
apictl get api-products -e production -k
apictl get apis -e production -k
```

To import the same product again, use `--update-api-product --update-apis` instead of `--import-apis`. Otherwise the import fails with a 409 error because the product already exists.

Expected Outcome

The CustomerLeasing API Product and the two APIs it uses are now in the production environment (PUBLISHED), and are visible in the production Developer Portal at `https://<APIM_HOST>:<PROD_HTTPS_PORT>/devportal`.

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